

Final Intern Presentation

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Introduction



- Mechanical engineering student at Texas A&M class of '27
- Throttle Control Engineer at Formula IC
- Intern for the R&D department under Chad Becht



The R&D team



First Project: Autonomously place the plastic overlay sheets into the ShopBot and its magnets

Why was it scrapped?

- Requires high dexterity
- Too much variation
- Unable to increase throughput



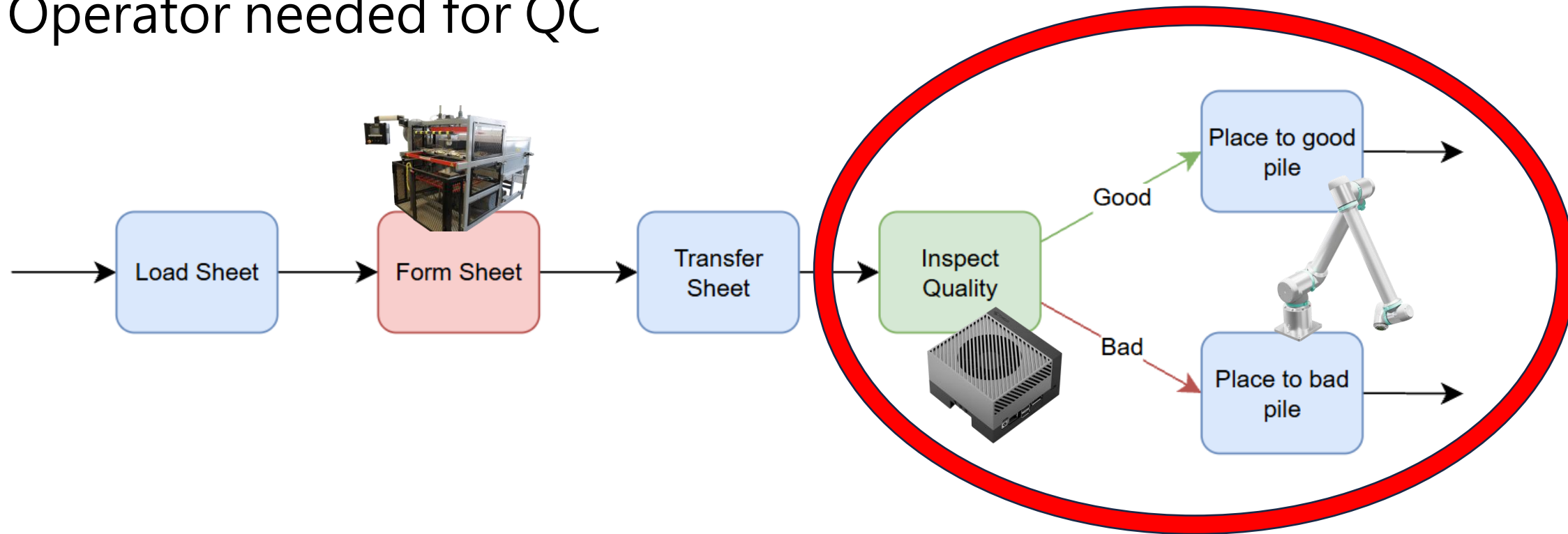
As the R&D team it's important to know when to pivot

Internship Project



Autonomously tend to the plastic overlay former

- Repetitive process
- Low dexterity
- Operator needed for QC



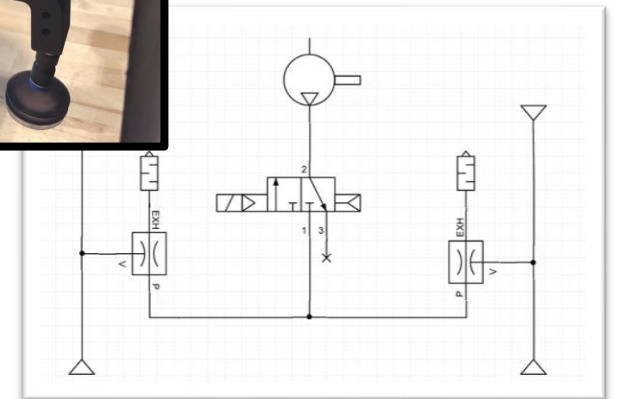
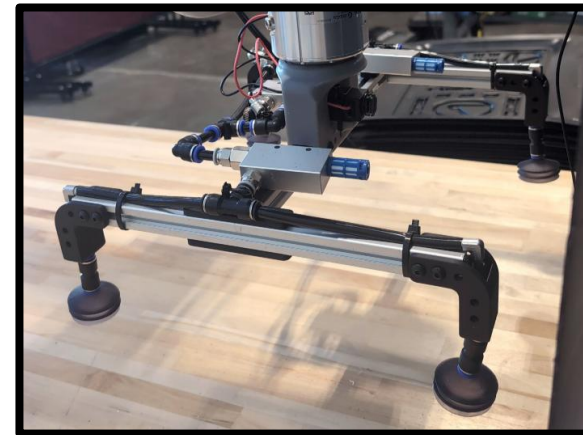
Deploying RobCo Robot Arm



- Installed the robot
 - Mounting
 - Electrical Wiring
 - Network connection
 - Repair \$\$\$
- Created a custom EOAT
 - Pneumatic Circuit
 - Custom suction cups



-- Mighty Mertha --

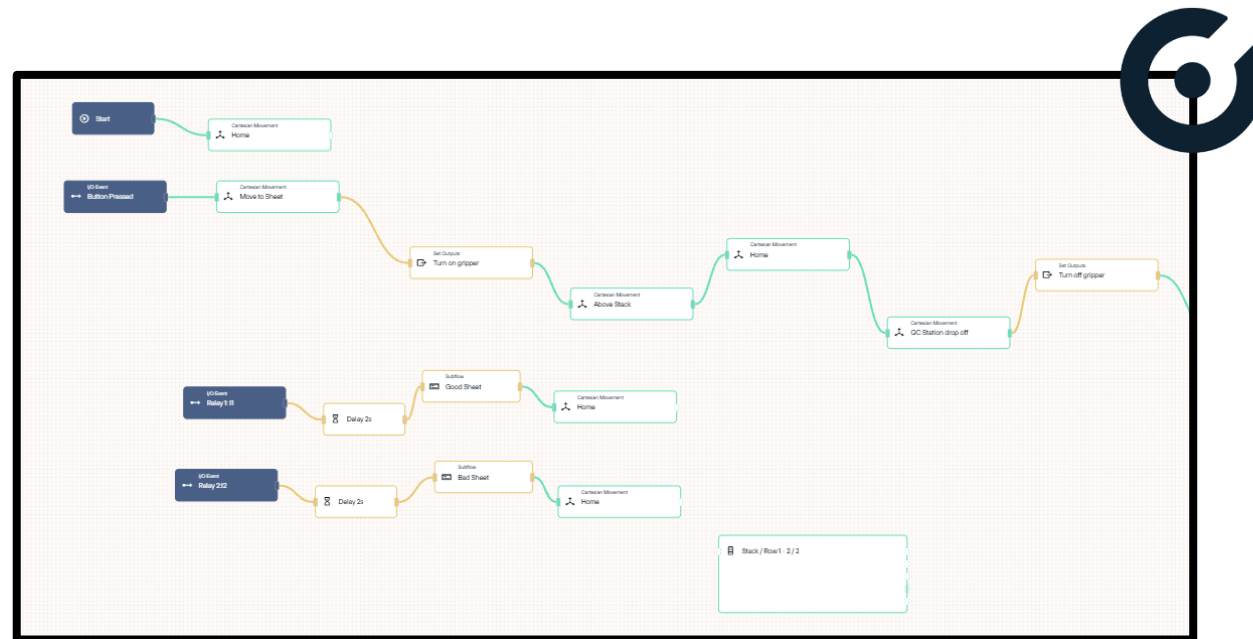


Programming the robot



RobFlow: Click and drag software that lets users create robot workflow

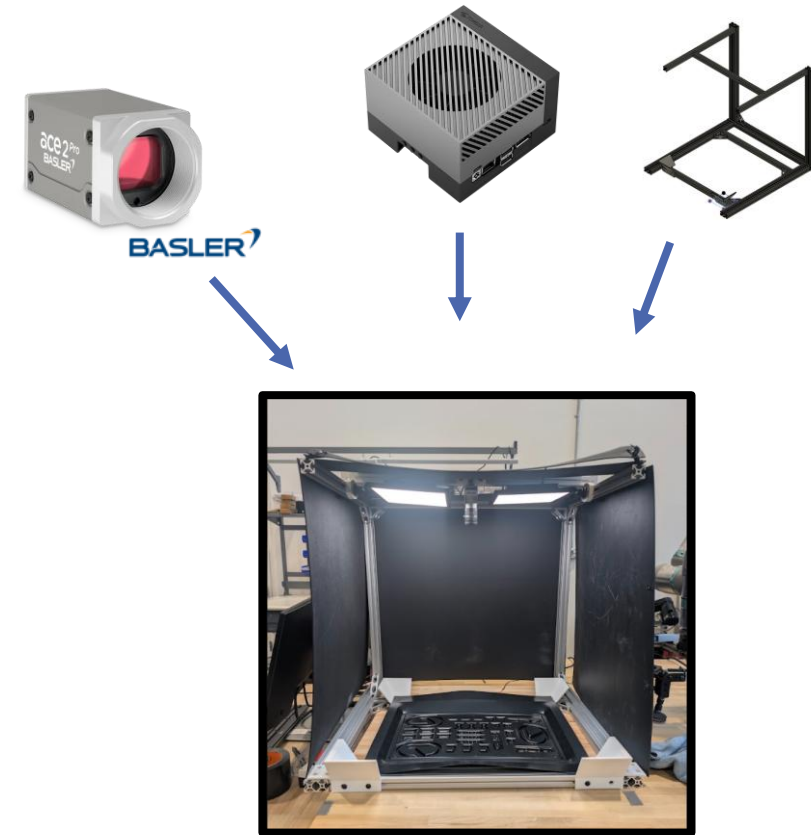
Programmed the main workflow whereby the robot places sheets into the QC Station and sorts sheets into good and bad piles



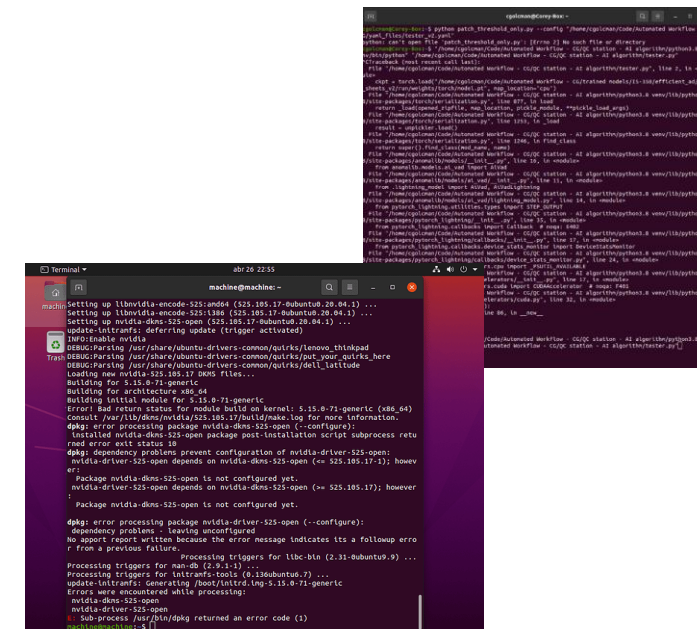
Constructing the QC Station



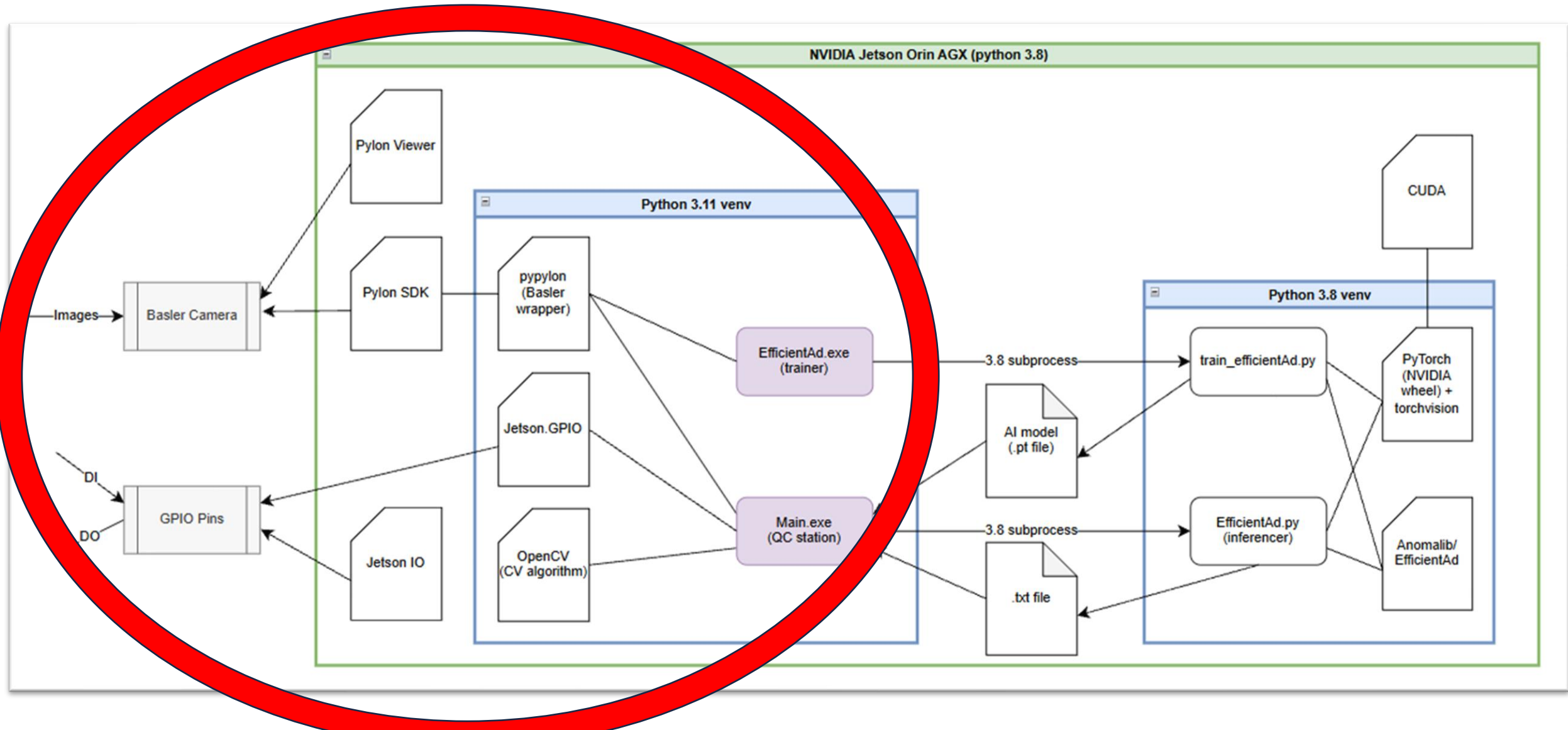
- Computer: NVIDIA Jetson Orin AGX
- Structure: T-slot aluminum frame
- Camera & Lense: Basler ace + Lense
 - High resolution, wide FOV and monochrome
- Lighting: Diffuse lighting panels
 - Dark curtains to block ambient lighting



- NVIDIA computer don't just work outside the box
- Integrating the NVIDIA ecosystem into the hands of ASG



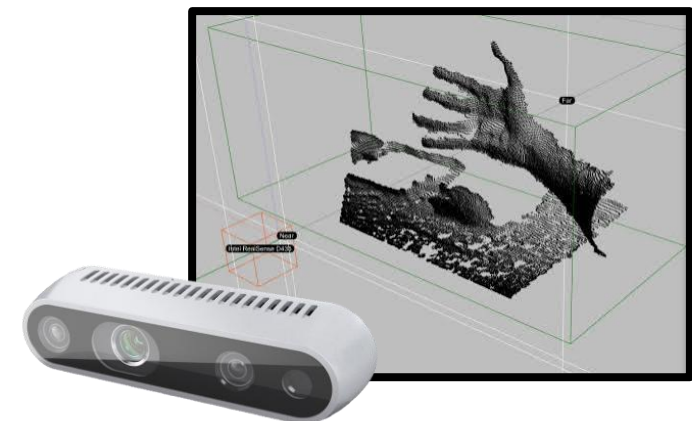
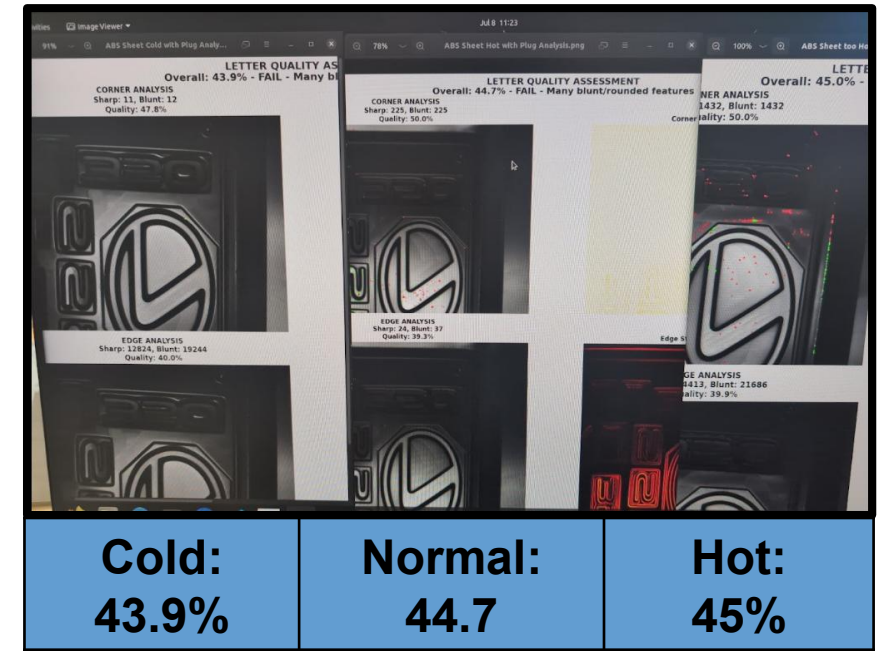
Software Stack



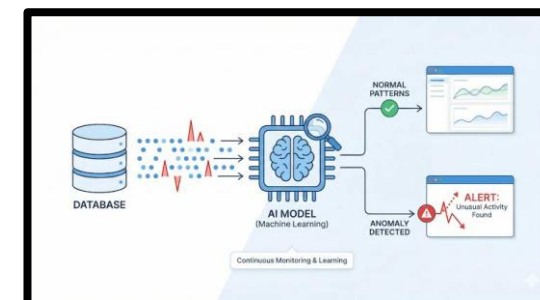
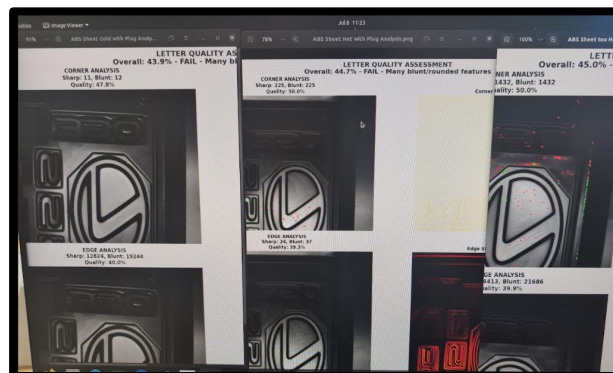
Classical CV Algorithms – another pivot



- Classical Computer Vision: mathematical algorithms to analyze part quality
- Slanted Edge Method(ISO 12233):
 - Industry standard algorithm for measuring edge profiles
- Infra red stereo camera:
 - Satin black sheets absorb the IR
 - Lacking in resolution

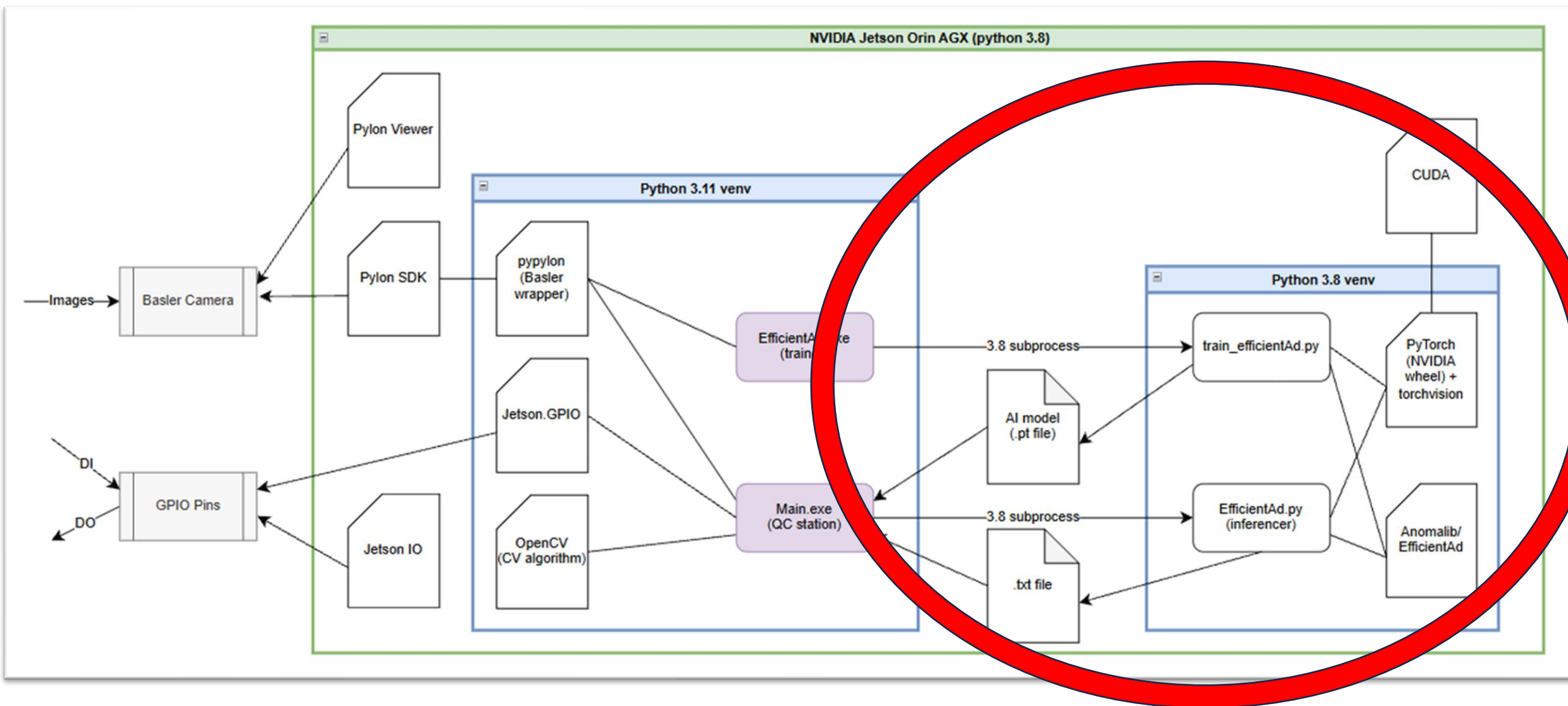


Computer Vision Algorithms



	1. Classical CV Algorithm	2. AI Algorithm
Pros	• Fast analysis times (~3s) • No GPU needed (run on any machine) • No training data required	• Adapts to natural variations in lighting, texture and part orientation • Can flag any type of variation
Cons	• Requires a unique algorithm per defect • Sensitive to variation in part orientation, lighting changes, etc.	• Requires GPU/edge compute (NVIDIA Jetson) • Requires a large training dataset (≥ 200 images) • Longer analysis time (~9s)

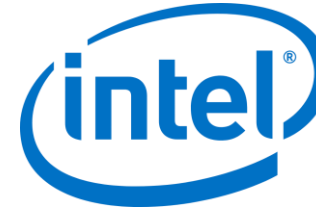
Software Stack



EfficientAd

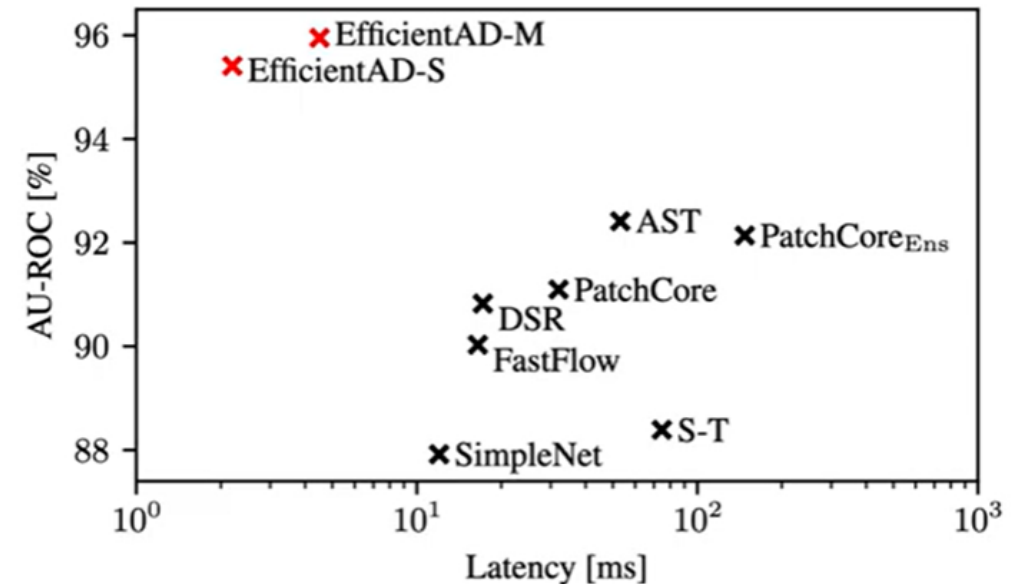


- Anomalib: Open-source deep learning library focused on visual anomaly detection from Intel
- EfficientAd: Most advanced anomaly detection AI (for our hardware)



	Small model	Medium model
Acc	95.4 %	96 %
Latency	2.2 ms	4.5 ms

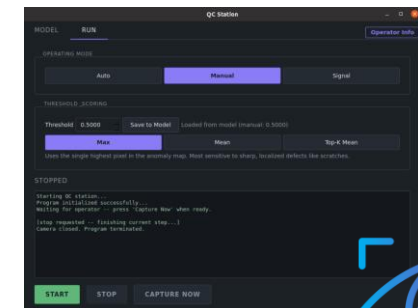
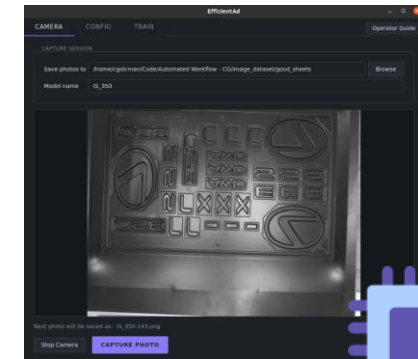
assumes perfect lab conditions 254p 3,000 images



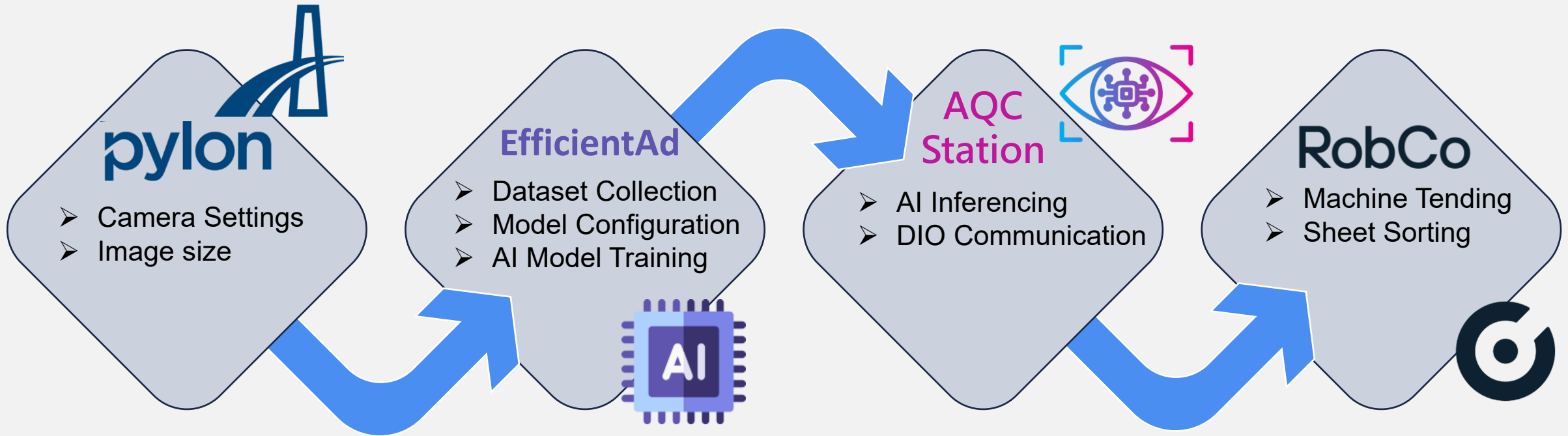
Custom Software



- EfficientAd.exe:
 1. Creates image dataset
 2. Configures a training instance
 3. Trains and creates an AI model
- QC Station.exe: Executes the inference model
 - Deployable for troubleshooting and for on the floor operation



Automated QC Workflow



Results & Impact



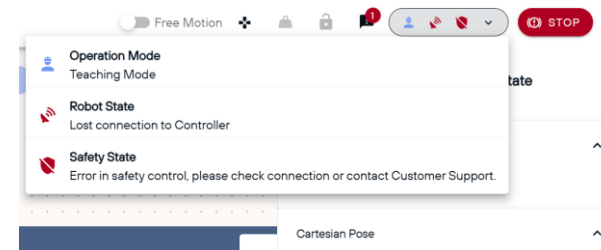
- A robust proof of concept that AI, machine vision and robotic technologies can be implemented in house
- Pioneering automation solutions that do not yet exist in the market giving ASG a competitive advantage
- Laid the foundation for future deployment on the shop floor and opened the doors to new possibilities

Challenges



- Many unfamiliar engineering concepts
- Long company password ***** 😊
- Getting my code working; many late nights debugging
- Staying focused: Other interns were distracting, lots of construction
- Company Firewall blocking all my software tools and robot
(shoutout Alexis for the help)
- Limited documentation from RobCo meaning I had to self teach and it led to a broken robot 😞

```
cat: c1: No such file or directory
cat: c2: No such file or directory
cat: c3: No such file or directory
cat: c4: No such file or directory
cat: c5: No such file or directory
cat: c6: No such file or directory
cat: c7: No such file or directory
cat: c8: No such file or directory
cat: c9: No such file or directory
duke@nukem ~$ stat nofile
stat: cannot stat 'nofile': No such file or directory
duke@nukem ~$ apt-cache show ffo
N: Unable to locate package ffo
E: No packages found
duke@nukem ~$ cp 55 66
cp: cannot stat '55': No such file or directory
duke@nukem ~$
```



Lessons Learned



- Lots of new engineering concepts:
 - Vacuum ejection/pneumatics
 - SLA 3D printing
 - Optics (cameras, lenses and lighting)
 - Machine vision
 - AI anomaly detection
 - Industrial Automation
- AI is a powerful tool that can amplify one's skill but must be used skeptically and analytically
- R&D is like a maze; one iteration may be a dead end, another may be the key to the finish line



Biggest Takeaway



“Don’t let perfection get in the way of good enough”

- Chad, 2026

Where I see my Career

- Didn't know what to expect when I first came in
- Confident in my mechanical and now software skills
- Led to a passion in automation and machine vision
- Best internship I have ever had 😊

Demo time!